

# HENRY CYRIL

## Principal Systems Architect

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### PROFESSIONAL SUMMARY

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Principal Systems Architect with 19+ years of experience in OSS/BSS, Provisioning, 5G, Cloud-Native transformation, and large-scale network modernization for tier-1 U.S. telecom operators. Architect of intelligent, self-healing provisioning ecosystems and first-of-its-kind satellite-to-cell capabilities enabling nationwide messaging coverage. Inventor of **20+ patents** in System Design, AI/ML-based anomaly detection, outage prediction, dynamic auditing, and autonomous recovery. Proven leader in designing distributed systems that support **100M+ subscribers** with ultra-high reliability.

#### HIGHLIGHTS INCLUDE:

- Designed AI-powered provisioning analytics platform improving accuracy to **99.9%** and reducing manual recovery by **90%**.
- Led 5G NSA/SA enablement for **50M+ users**, establishing foundation for 5G Advanced and 6G evolution.
- Architected U.S.'s first **satellite-to-cell messaging** implementation with Starlink, including 911 emergency routing.
- Modernized provisioning to an active-active, self-healing system reducing fallout from **250K → 15K** monthly.
- Delivered BSS capabilities enabling Data Stash, Mobile Without Borders, Binge On, and Project 10 Million.

### KEY COMPETENCIES

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| <ul style="list-style-type: none"><li>• <b>Architecture &amp; System Design</b><ul style="list-style-type: none"><li>- System Design</li><li>- Architectural Design</li><li>- Microservices &amp; SOA</li><li>- Technical Architecture</li><li>- Capacity &amp; Performance Engineering</li></ul></li><li>• <b>Telecom OSS/BSS Expertise</b><ul style="list-style-type: none"><li>- Provisioning</li><li>- Order Management</li><li>- Rating, Charging &amp; Mediation</li><li>- Billing</li><li>- CRM / BSS Platforms</li></ul></li><li>• <b>Cloud, Distributed Systems &amp; DevOps</b><ul style="list-style-type: none"><li>- Kubernetes, Microservices</li><li>- Apache Kafka, Confluent,</li><li>- Cloud Applications</li><li>- Build Automation, CI/CD</li><li>- REST / RESTful APIs, SOAP</li><li>- Observability</li></ul></li></ul> | <ul style="list-style-type: none"><li>• <b>AI, ML &amp; Innovation</b><ul style="list-style-type: none"><li>- AI &amp; ML</li><li>- AI/ML for Anomaly Detection</li><li>- Predictive Outage Prevention</li><li>- Automated Root-Cause Analysis</li><li>- Innovation &amp; Research Design</li></ul></li><li>• <b>Business &amp; Product Skills</b><ul style="list-style-type: none"><li>- Application Development</li><li>- CRM Platforms</li><li>- IT/Network Operations</li><li>- Operations Improvement</li><li>- Product Roadmapping</li></ul></li><li>• <b>Programming &amp; Development</b><ul style="list-style-type: none"><li>- Core Java, Java, JavaScript</li><li>- JSON, C, C++, C#, .NET</li><li>- Objective-C, PHP</li><li>- Shell Scripting, Unix Scripting</li><li>- Oracle SQL, MySQL, XML</li></ul></li></ul> |
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## PROFESSIONAL HISTORY

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### Principal Engineer, Systems Architecture

Jan 2019 - Present

#### T-Mobile – Bellevue, WA

- Lead architect for nationwide modernization programs across provisioning, OSS/BSS, activation, network orchestration, and cloud-native transformation supporting 100M+ subscribers.
- Designed and delivered an AI-powered provisioning analytics platform improving accuracy to **99.9%**, reducing manual recovery by **90%**, and accelerating fault-isolation across network domains.
- Led 5G NSA/SA enablement architecture ensuring **50M+ customers** seamlessly transitioned to next-generation network capabilities with **99.9% provisioning accuracy**.
- Architected and deployed the **first satellite-to-cell messaging service in the U.S.** (T-Satellite + Starlink), developing provisioning logic, routing flows, and emergency 911 integration.
- Modernized legacy provisioning stack into an **active-active, self-healing** platform reducing fallout from **250K → 15K** monthly and improving MTTR by **50%**.
- Defined enterprise CI/CD architecture and cloud-native patterns, achieving **95% automation** and reducing deployment timelines from **5 days → 5 hours**.
- Key contributor and design authority for T-Mobile's OSS/BSS roadmap, providing governance across architecture reviews, vendor integrations, and reliability engineering.
- Recipient of multiple awards: **PEAK Award**, Universe Mover, EIT Extraordinaire, and Star Awards for leadership and innovation.

### Principal Analyst, Systems

Sep 2014 – Jan 2019

#### T-Mobile - Bellevue, WA

- Lead architect for T-Mobile's multi-year BSS Transformation including Billing, Rating, Charging, Mediation, CRM, and Order Management.
- Delivered architecture for flagship Un-carrier programs: **Data Stash**, **Mobile Without Borders**, **Binge On**, and **Project 10 Million**, enabling industry-first customer experience innovations.
- Designed real-time charging, roaming, and mediation flows ensuring accurate usage, rating, and policy control across millions of subscriber transactions.
- Drove end-to-end design for prepaid/postpaid unification and multi-segment migrations, ensuring consistent provisioning and billing across product lines.
- Led design reviews with vendors (Ericsson, E///, others), validating HLD/LLD deliverables, network configurations, and integration logic.
- Supported performance tuning, issue triage, defect analysis, and architecture harmonization across Billing, EPC, Charging, and CRM systems.
- Awarded Universe Mover, EIT Extraordinaire, and multiple Star recognitions.

## BSS Specialist

Mar 2013 – Sep 2014

### Ericsson - Bangalore, India

- Designed and optimized BSS nodes for Telstra (Australia) including charging, billing, mediation, CRM, and partner settlement systems.
- Implemented BI/KPI dashboards, reconciliation engines, and revenue-assurance workflows improving operational accuracy and reporting transparency.
- Developed automation for performance monitoring, contractual KPI compliance, and usage analytics using Pentaho, SQL, and Java-based frameworks.

## Team Lead, GNSC Portal

Jul 2011 – Mar 2013

### Nokia Siemens Networks – Chennai, India

- Led development of KPI dashboards, RCA modules, network analytics tools, and 65+ automation utilities used by 4,000+ NSN employees worldwide.
- Designed architecture for customer experience management (CEM), PM/FM analytics, SLA/OLA dashboards, and multi-system drill-down diagnostics.
- Integrated OSS, CS-Core, Packet Core, BSC, HLR, and Remedy systems enabling unified operational views for global support teams.

## VAS Specialist

Nov 2008 - Jul 2011

### Nokia Siemens Networks – Chennai, India

- Designed CRM workflows, charging logic, IVR integrations, provisioning flows, multi-tenant architectures, and MNP (Mobile Number Portability) processes for MVNO clients.
- Built interface layers for CRM, Billing, Top-Up, Voucher, Loyalty, IVR, and mediation systems using SOA, web services, and data transformation frameworks.

## Assistant Systems Engineer

Oct 2006 - Nov 2008

### Tata Consultancy Services

- Developed IMS/SIP-based VoIP client for Nokia Siemens Networks supporting voice/video calling, conferencing, jitter buffering, and codec integration (G711, G729).
- Built middleware integrations for British Telecom using IBM Message Broker, MQ, SOAP, and REST services enabling real-time B2B data flows.

## SELECTED MAJOR PROJECTS & INNOVATIONS

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### T-Satellite Integration (Starlink + T-Mobile Satellite-to-Cell Messaging)

T-Mobile | Principal Systems Architect | 2024 – 2025

- Architected the **first satellite-to-cell messaging service in the U.S.**, enabling coverage in remote/no-signal regions.
- Designed provisioning logic allowing devices to automatically switch between terrestrial and satellite networks.
- Developed emergency 911 satellite message routing meeting FCC compliance requirements.
- Built asynchronous APIs integrating T-Mobile OSS/BSS with Starlink satellite controllers.
- Executed a controlled, phased enablement strategy ensuring zero customer-impact during rollout.

## AI-Powered Network Provisioning Analytics Platform

T-Mobile | Principal Systems Architect | 2023 – 2025

- Architected an enterprise-scale AI/ML analytics platform providing real-time anomaly detection, predictive outage prevention, and automated recovery across provisioning, billing, and network domains.
- Designed self-healing workflows reducing manual recovery by 90% and improving provisioning accuracy to 99.9%.
- Implemented AI-driven dynamic auditing improving cross-domain data accuracy from 98.5% → 99.9%.
- Built predictive models for capacity planning, outage mitigation, and performance optimization supporting 100M+ subscribers.
- Delivered unified observability dashboards enabling real-time reliability insights for leadership and operations.

## Provisioning Infrastructure Modernization (Active-Active, Self-Healing)

T-Mobile | Principal Systems Architect | 2024 – 2025

- Modernized legacy provisioning into a fully redundant **active-active architecture** achieving continuous availability.
- Redesigned deterministic transaction workflows improving data consistency and platform stability.
- Reduced fallout from **250K → 15K** monthly through intelligent recovery and dynamic traffic shaping.
- Built a controlled migration strategy allowing real-time validation and zero customer-impact cutovers.
- Achieved **99.98% provisioning success** across millions of monthly activations.

## Cloud-Native Transformation & CI/CD Orchestration Platform

T-Mobile | Principal Architect | 2022 – 2023

- Led end-to-end migration of mission-critical provisioning and network query workloads from VMs to Kubernetes microservices.
- Designed CI/CD architecture achieving **95% automation** and reducing deployment cycles from **5 days → 5 hours**.
- Introduced geo-redundant failover and containerized scaling improving availability from **75% → 99%**.
- Established North Star architecture, interface specifications, and migration strategy using the Strangler pattern.
- Delivered 50% hardware/resource savings through container orchestration and dynamic scaling.

## 5G Enablement & Provisioning Architecture (NSA + SA)

T-Mobile | Principal Architect | 2020 – 2023

- Designed end-to-end provisioning architecture for **5G NSA and SA** enabling 5G capabilities for **50M+ customer accounts**.
- Implemented automated enablement embedding 5G profiles into all provisioning flows, reducing migration risk.
- Delivered **99.9% provisioning accuracy**, reducing activation failures by 40% and improving operational efficiency.
- Created lifecycle processes aligning OSS/BSS, Core Network, and Device Engineering for 5G adoption.
- Established foundation for 5G Advanced and future 6G roadmap.

## Cross-Provisioning & Streaming Conversion (Sprint → T-Mobile Merger)

T-Mobile | Principal Architect | 2020 – 2022

- Architected cross-provisioning framework enabling **10M+ customer migrations** with zero downtime during the Sprint–T-Mobile merger.
- Designed asynchronous APIs, queuing layers, and automated reprovisioning flows ensuring accuracy and real-time synchronization.
- Migrated multiple device types (voice lines, IoT, hotspots, home internet) without impacting active services.
- Ensured full compliance for Lifeline/Assurance Wireless customers with preserved benefits and eligibility.
- Reduced manual migration effort by **80%** and improved throughput by **40%**.

## T-Mobile BSS Transformation (Billing, Rating, Charging, CRM)

T-Mobile | Principal Analyst / Architect | 2014 – 2019

- Led architecture for nationwide BSS modernization enabling Un-carrier programs including **Data Stash, Mobile Without Borders, Binge On, and Project 10 Million**.
- Designed rating, charging, mediation, billing, and CRM workflows serving tens of millions of customers.
- Implemented modular rating frameworks reducing time-to-market by **70%** for new product launches.
- Ensured real-time synchronization across CRM, billing, charging, EPC, and provisioning systems.
- Contributed to multiple J.D. Power award-winning customer experience improvements.

## T-Life App — Real-Time Network Diagnostics & Self-Care

T-Mobile | Principal Systems Architect | 2025 – Present

- Led architecture for T-Life's real-time diagnostic engine providing customers live visibility into network status and service issues.
- Designed autonomous validation framework testing thousands of rate-plan/device combinations to ensure diagnostic accuracy.
- Improved diagnostic accuracy by **35%** and reduced validation cycles from **15 minutes → 30 seconds**.
- Delivered actionable analytics reducing customer-care dependency and enabling 20–25% reduction in support calls.
- Elevated T-Life as a next-generation digital self-care platform with nationwide adoption.

## PATENTS & INNOVATION

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### AI/ML for Reliability & Outage Prediction

- Network Provisioning Engine — AI/ML–Based Anomaly Detection & Outage Prediction (*Published: US 20250337641*)
- AI/ML Recommendation Engine for Proactive Resource Allocation in Provisioning Systems (*Published: US 20250344233*)
- Intelligent AI/ML–driven failover decision-making
- Proactive outage prediction based on deployment logs and automation testing
- Intelligent heartbeat logic during network element outage events
- ML-based intelligent traffic routing for optimized provisioning & network path selection

### Platform Optimization & Transaction Intelligence

- Dynamic real-time provisioning validation utility
- Network provisioning catalog comparison engine
- Dynamic real-time network audit function
- Realtime provisioning status notifications
- Transaction serialization engine & common POH serialization
- Dynamic queueing in provisioning systems
- Seamless provisioning transaction queueing during network element degradation/outages
- Dynamic Provisioning Reconciliation (*Published: US 20220295351*)
- Dynamic In-Memory Cache Update for Provisioning Transactions (*Published: US 20230188956*)

### Revenue Assurance, Leak Prevention & ML-Based Recommendations

- AI/ML-based real-time revenue leak detection caused by provisioning/charging mismatch
- AI/ML-based dynamic rate plan / add-on recommendation engine
- Reduce revenue leak during natural disaster / emergency suspension scenarios

### Customer Diagnostics, Testing Automation & Self-Healing

- Self-Care Network Diagnostics & Automated Resolution

## EDUCATION

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- **Master of Business Administration (MBA), Systems** - 2009  
Bharathidasan University
- **Bachelor of Engineering (BE), Electronics & Communication** - 2006  
Anna University